Asssignment 3

Astrophysics 509

DATE March 2, 2026

Jaynard Kigas (30173694)

The University of Calgary

Contents

Question 1 **1**

Question 2 **1**

Question 2.a **1**

Question 2.b **2**

Question 3 **2**

Question 3.a **2**

Question 3.b **3**

Question 3.c **3**

Question 4 **3**

Question 5 **5**

Question 1

For a white dwarf of density $1 \times 10^6\,\mathrm{g/cm^3}$ consisting of only $^{12}\mathrm{C}$, estimate the degeneracy pressure and compare it with the thermal pressure of a gas of temperature $1 \times 10^7\,\mathrm{K}$.

The degeneracy pressure is given as:

$$P \approx 0.8\hbar c\left(\frac{Z}{A}\right)^{4/3}\left(\frac{\rho}{m_p}\right)^{4/3} \tag{1.1}$$

Where Z is the number of protons in each species (we only have carbon-12), A is the number of nucleons (12), ρ the density, and $m_p = 1.673 \times 10^{-24}\,\mathrm{g}$.

$$P \approx 0.8\left(\frac{6.626 \times 10^{-27}\,\mathrm{erg\ s}}{2\pi}\right)(2.998 \times 10^{10}\,\mathrm{cm/s})\left(\frac{6}{12}\right)^{4/3}\left(\frac{1 \times 10^6\,\mathrm{g/cm^3}}{1.673 \times 10^{-24}\,\mathrm{g}}\right)^{4/3} \tag{1.2}$$

$$\boxed{P \approx 5.05 \times 10^{22}\,\mathrm{Ba}} \tag{1.3}$$

The thermal pressure is then simply the ideal gas pressure:

$$PV = Nk_BT \tag{1.4}$$

$$P = \frac{\rho k_B T}{12 m_p} = \frac{(1 \times 10^6\,\mathrm{g/cm^3})(1.381 \times 10^{-16}\,\mathrm{erg/K})(1 \times 10^7\,\mathrm{K})}{12(1.673 \times 10^{-24}\,\mathrm{g})} \tag{1.5}$$

$$\boxed{P = 6.88 \times 10^{19}\,\mathrm{Ba}} \tag{1.6}$$

So the pressures differ by about a factor of 100.

Question 2

White Dwarf III:

Question 2.a

What is the thermal energy stored in a white dwarf of temperature $T = 1 \times 10^7\,\mathrm{K}$ and a mass of $1\,\mathrm{M}_\odot$?

The thermal energy is given by:

$$U = \frac{3}{2}Nk_BT = \frac{3}{2}\frac{1\,\mathrm{M}_\odot}{12\,\mathrm{m_p}}k_BT \tag{2.1}$$

$$U = \frac{3}{2}\frac{1.989 \times 10^{33}\,\mathrm{g}}{12(1.673 \times 10^{-24}\,\mathrm{g})}(1.381 \times 10^{-16}\,\mathrm{erg/K})(1 \times 10^7\,\mathrm{K}) \tag{2.2}$$

$$\boxed{U = 2.05 \times 10^{47}\,\mathrm{erg}} \tag{2.3}$$

Question 2.b

Assuming that the white dwarf radiates like a black body of temperature $T = 1 \times 10^4\,\mathrm{K}$, estimate its lifetime as a luminous object.

Let's find the luminosity.

$$L = 4\pi R^2 \sigma T^4 \tag{2.4}$$

Where T is the black body temperature. The age estimate is then how long this luminosity runs through the internal energy as the white dwarf does not produce any more energy. let's now find the radius:

$$R = \frac{\hbar^2}{G m_e}\left(\frac{3Z}{4\pi A m_p}\right)^{5/3} M^{-1/3} \tag{2.5}$$

$$R = \frac{(6.626 \times 10^{-27}\,\mathrm{erg\ s})^2}{4\pi^2 (6.673 \times 10^{-8}\,\mathrm{dyne\ cm^2/g^2})(9.110 \times 10^{-28}\,\mathrm{g})}\left(\frac{3(6)}{4\pi(12)(1.673 \times 10^{-24}\,\mathrm{g})}\right)^{5/3}\left(1.989 \times 10^{33}\,\mathrm{g}\right)^{-1/3} \tag{2.6}$$

$$R = 1.79 \times 10^7\,\mathrm{cm} \tag{2.7}$$

Okay now we're just supposed to throw this radius out and give the more accurate answer derived from the figure in the textbook.

$$R = 0.6 \times 10^9\,\mathrm{cm} \tag{2.8}$$

The luminosity is then:

$$L = 4\pi(0.6 \times 10^9\,\mathrm{cm})^2(5.670 \times 10^{-5}\,\mathrm{erg/cm^2/K^4/s})(1 \times 10^4\,\mathrm{K})^4$$

$$L = 2.57 \times 10^{30}\,\mathrm{erg/s} \tag{2.9}$$

So now the luminous time is:

$$\frac{U}{L} = \frac{2.05 \times 10^{47}\,\mathrm{erg}}{2.57 \times 10^{30}\,\mathrm{erg/s}} = \boxed{7.98 \times 10^{16}\,\mathrm{s}} \tag{2.10}$$

This is in the billions of years.

Question 3

Type Ia SN:

Question 3.a

Given a total radiated energy of $1 \times 10^{50}\,\mathrm{erg}$, and that the energy entirely comes from Ni to Fe decay, determine the amount of Ni mass (in units of $M_\odot$) required.

We can find this by finding the energy release per unit mass. the energy released in the chain is about:

$$1.1 \times 10^{50}\,\mathrm{erg} \rightarrow 0.6\,\mathrm{M}_\odot \tag{3.1}$$

$$1 \times 10^{50}\,\mathrm{erg} \cdot \frac{0.6\,\mathrm{M}_\odot}{1.1 \times 10^{50}\,\mathrm{erg}} = 0.545\,\mathrm{M}_\odot \tag{3.2}$$

Question 3.b

How long does it take for the luminosity to drop by a factor of 10 during the phase that Co decay dominates the luminosity?

The halflife of Cobalt decay is given as $\tau = 77.7\,\text{days}$. The luminosity relation between this is:

$$L = L_0 e^{-\lambda t} \tag{3.3}$$

Where $\lambda = \ln(2)/\tau$. We have:

$$0.1 L_0 = L_0 e^{-\ln(2)t/\tau} \tag{3.4}$$

$$\ln(0.1) = -\ln(2) t\tau \tag{3.5}$$

$$-\frac{\ln(0.1)}{\ln(2)}\tau = t \tag{3.6}$$

$$t = 3.32\tau = 258\,\text{days} \tag{3.7}$$

So it takes 258 days lol.

Question 3.c

Does the luminosity drop more quickly or more slowly when the supernova becomes optically thin to Co decay radiation?

The luminosity drops more quickly since energy, in the form of high energy rays, escapes the system faster (no internal heat source from scattering).

Question 4

Calculate the luminosity for a supernova with initial Ni mass of $0.1\,\text{M}_\odot$, at 100d after explosion, including both ^{56}Ni and ^{56}Co decay.

The decay of nickle has a half-life of $\tau = 6.1\,\text{days}$ and the decay of cobalt has a decay $\tau = 77.7\,\text{days}$. We have the luminosity as:

$$L = \epsilon \frac{\mathrm{d}M}{\mathrm{d}t} = \epsilon_m m_{nuc} \frac{\mathrm{d}N}{\mathrm{d}t} \tag{4.1}$$

We also have our decay chains (nickle - cobalt - iron) where:

$$\frac{\mathrm{d}N_{Ni}}{\mathrm{d}t} = -\lambda_{Ni} N_{N_i} \tag{4.2}$$

$$\frac{\mathrm{d}N_{Co}}{\mathrm{d}t} = -\lambda_{Co} N_{Co} + \left(-\frac{\mathrm{d}N_{Ni}}{\mathrm{d}t}\right) \tag{4.3}$$

Let's solve the first equation (it's trivial):

$$N_{N_i} = N_{Ni,0} e^{-\lambda_{Ni} t} \tag{4.4}$$

Let's insert this solution into 4.2:

$$\frac{\mathrm{d}N_{Co}}{\mathrm{d}t} + \lambda_{Co} N_{Co} = \left(-\frac{\mathrm{d}N_{Ni}}{\mathrm{d}t}\right) \tag{4.5}$$

$$\frac{\mathrm{d}N_{Co}}{\mathrm{d}t} + \lambda_{Co} N_{Co} = \lambda_{Ni} N_{Ni,0} e^{-\lambda_{Ni} t} \tag{4.6}$$

This is a differential equation to solve:

$$\int \lambda_{Co}\,\mathrm{d}t = \lambda_{Co} t \tag{4.7}$$

$$u = e^{\lambda_{Co} t} \tag{4.8}$$

$$N_{Co} e^{\lambda_{Co} t} = \int \lambda_{Ni} N_{Ni,0} e^{-\lambda_{Ni} t} e^{\lambda_{Co} t}\,\mathrm{d}t \tag{4.9}$$

$$N_{Co} e^{\lambda_{Co} t} = \lambda_{Ni} N_{Ni,0} \frac{1}{\lambda_{Co} - \lambda_{Ni}} e^{t(\lambda_{Co} - \lambda_{Ni})} + C \tag{4.10}$$

Since $N_{Co}(0) = 0$, We have

$$C = -\frac{\lambda_{Ni} N_{Ni,0}}{\lambda_{Co} - \lambda_{Ni}} \tag{4.11}$$

So finally

$$N_{Co} = \frac{\lambda_{Ni} N_{Ni,0}}{\lambda_{Co} - \lambda_{Ni}} e^{-t\lambda_{Ni}} - \frac{\lambda_{Ni} N_{Ni,0}}{\lambda_{Co} - \lambda_{Ni}} e^{-t\lambda_{Co}} \tag{4.12}$$

$$N_{Co} = \frac{\lambda_{Ni} N_{Ni,0}}{\lambda_{Co} - \lambda_{Ni}} \left(e^{-t\lambda_{Ni}} - e^{-t\lambda_{Co}}\right) \tag{4.13}$$

So our full derivative is then:

$$\frac{\mathrm{d}N}{\mathrm{d}t} = \frac{\lambda_{Co}\lambda_{Ni} N_{Ni,0}}{\lambda_{Co} - \lambda_{Ni}} \left(e^{-t\lambda_{Ni}} - e^{-t\lambda_{Co}}\right) + \lambda_{Ni} N_{Ni,0} e^{-\lambda_{Ni} t} \tag{4.14}$$

Which is fully positive since we are plugging straight into our luminosity equation (where losing particles is a gain in luminosity). We now have:

$$L = \epsilon_m m_{nuc} \left(\frac{\mathrm{d}N}{\mathrm{d}t}\right) = \epsilon_{Co} m_{Co} \left(\frac{\lambda_{Co}\lambda_{Ni} N_{Ni,0}}{\lambda_{Co} - \lambda_{Ni}} \left(e^{-t\lambda_{Ni}} - e^{-t\lambda_{Co}}\right)\right) + \epsilon_{Ni} m_{Ni} \lambda_{Ni} N_{Ni,0} e^{-\lambda_{Ni} t} \tag{4.15}$$

This can be reduced since $m_{nuc} N_0 = M_0$:3

$$L = \epsilon_{Co} \left(\frac{\lambda_{Co}\lambda_{Ni} M_0}{\lambda_{Co} - \lambda_{Ni}} \left(e^{-t\lambda_{Ni}} - e^{-t\lambda_{Co}}\right)\right) + \epsilon_{Ni} M_0 \lambda_{Ni} e^{-\lambda_{Ni} t} \tag{4.16}$$

We have $\epsilon_{Co} = 6.4 \times 10^{16}\,\mathrm{erg/g}, \epsilon_{Ni} = 3 \times 10^{16}\,\mathrm{erg/g}$. This luminosity at 100 days is then:

$$L = \epsilon_{Co} \left(\frac{\lambda_{Co}\lambda_{Ni} M_0}{\lambda_{Co} - \lambda_{Ni}} \left(e^{-t\lambda_{Ni}} - e^{-t\lambda_{Co}}\right)\right) + \epsilon_{Ni} M_0 \lambda_{Ni} e^{-\lambda_{Ni} t} \tag{4.17}$$

$$L(100) = (6.4\times10^{16}\,\mathrm{erg/g}) \left(-\ln(2) \frac{1.989 \times 10^{32}\,\mathrm{g}}{(6.1)(77.7)(1/77.7 - 1/6.1)}\right) \left(e^{-100\ln(2)/6.1} - e^{-100\ln(2)/77.7}\right) + (3\times10^{16}\,\mathrm{erg/g})(1.989\times10^{32}\,\mathrm{g} \tag{4.18}$$

Oh wow it goes off the screen. Let's condense it:

$$L(100)_{Co} = (6.4 \times 10^{16}\,\mathrm{erg/g}) \left(\ln(2) \frac{1.989 \times 10^{32}\,\mathrm{g}}{(6.1)(77.7)(1/77.7 - 1/6.1)}\right) \left(e^{-100\ln(2)/6.1} - e^{-100\ln(2)/77.7}\right) \tag{4.19}$$

$$L(100)_{Ni} = (3 \times 10^{16}\,\mathrm{erg/g})(1.989 \times 10^{32}\,\mathrm{g})(\ln(2)/6.1)e^{-100\ln(2)/6.1} \tag{4.20}$$

I also ommitted writing the unit for time (day) explicitly because its encumbersome. Summing these together we get:

$$L(100) = 5.0499 \times 10^{46}\,\mathrm{erg/day} + 7.876 \times 10^{42}\,\mathrm{erg/day} = \boxed{5.05 \times 10^{46}\,\mathrm{erg/day}} \tag{4.21}$$

Or in seconds we have it:

$$L = 5.85 \times 10^{41}\,\mathrm{erg/s} \tag{4.22}$$

Question 5

Calculate the mass-radius relation for white dwarfs, as follows: Use the exact expression for the Fermi momentum and the non-relativistic expression for pressure of a degenerate electron gas. Assume the white dwarf is pure ionized 12C. Integrate the equation of hydrostatic equilibrium for a constant density white dwarf to get the central pressure and hence the mass radius relation. Give your final result as R as a function of $(M/M_\odot)$ with a numerical coefficient.

The fermi momentum is:

$$p_f = \left(\frac{3h^3 n_e}{8\pi}\right)^{1/3} \tag{5.1}$$

This density of the electron is related to the nuclei density since our star is fully carbon-12

$$n_e = \frac{Z}{A}\frac{\rho}{m_p} \tag{5.2}$$

Our fermi momentum is then:

$$p_f = \left(\frac{3h^3}{8\pi}\frac{Z\rho}{Am_p}\right)^{1/3} \tag{5.3}$$

We will use this in the pressure expression from momentum.

$$P = \frac{8\pi}{15h^3 m_e}p_F^5 = \frac{8\pi}{15h^3 m_e}\left(\frac{3h^3}{8\pi}\frac{Z\rho}{Am_p}\right)^{5/3} \tag{5.4}$$

Where the hydrostatic equilibrium relation is as follows:

$$\frac{\mathrm{d}P}{\mathrm{d}r} = -\rho\frac{Gm(r)}{r^2} \tag{5.5}$$

We can then integrate this over P for the LHS and over r for the RHS. Our bounds of integration will be from the core (Where we have a central pressure P_c and to the stellar radius where pressure is zero)

$$P\Big|_{P_c}^{0} = -G\rho\int_0^R \frac{m(r)}{r^2}\,\mathrm{d}r \tag{5.6}$$

Let's deal with this integral on the RHS:

$$\int_0^R \frac{m(r)}{r^2}\,\mathrm{d}r = \int_0^R \frac{\rho(4/3)\pi r^3}{r^2}\,\mathrm{d}r = (2/3)\rho\pi R^2 \tag{5.7}$$

So the total RHS is then:

$$-(2/3)G\rho^2\pi R^2 \tag{5.8}$$

We then have:

$$P_c = \frac{2G\rho^2\pi R^2}{3} \tag{5.9}$$

Let's equate this to our pressure derived from fermi momentum:

$$\frac{8\pi}{15h^3m_e}\left(\frac{3h^3}{8\pi}\frac{Z\rho}{Am_p}\right)^{5/3} = \frac{2G\rho^2\pi R^2}{3} \tag{5.10}$$

Let's simplify math by considering that we have constant density (a requirement since we integrated over r) and that the density is the total mass M divided by the volume.

$$\frac{8\pi}{15h^3m_e}\left(\frac{3h^3}{8\pi}\frac{Z}{Am_p}\frac{M}{(4/3)\pi R^3}\right)^{5/3} = \frac{2G\pi R^2}{3}\frac{M^2}{(16/9)\pi^2R^6} \tag{5.11}$$

Now we begin the arduous journey of solving for R.

$$\frac{8\pi h^5}{15h^3m_eR^5}\left(\frac{9Z}{32\pi^2Am_p}\right)^{5/3}M^{5/3} = \frac{3GM^2}{8\pi R^4} \tag{5.12}$$

$$R = \frac{64\pi^2h^2}{45Gm_e}\left(\frac{9Z}{32\pi^2Am_p}\right)^{5/3}M^{-1/3} \tag{5.13}$$

So now we have R as a function of M with a constant out front. This is going to be hell to compute but it is (for carbon-12):

$$R = 3.60\times10^{19}\,\mathrm{cmg}^{1/3}M^{-1/3} \tag{5.14}$$

However, to get this in units of mass fraction of solar mass, we must divide by $M_\odot^{-1/3}$. This gives the constant as:

$$3.60\times10^{19}\,\mathrm{cmg}^{1/3}\cdot\frac{(1.989\times10^{33}\,\mathrm{g})^{-1/3}}{M_\odot^{-1/3}} = 2.86\times10^{8}\,\mathrm{cmM}_\odot^{-1/3} \tag{5.15}$$

So finally

$$\boxed{R = 2.86\times10^{8}\,\mathrm{cmM}_\odot^{1/3}M^{-1/3}} \tag{5.16}$$

Where M is expressed as a fraction of solar mass.

References